

Robust Auto-Tuning Control of a Delivery Quadcopter with Motor Faults, Mass and Inertia Estimation

M.A. Mafaz* N. Horri** Q. Lu*** M. England*

* *Centre for Computational Science and Mathematical Modelling, Coventry University, Coventry, UK (e-mail: mohamedmam@uni.coventry.ac.uk, Matthew.England@coventry.ac.uk)*

** *School of Engineering, University of Leicester, Leicester, UK (e-mail: nmh22@leicester.ac.uk)*

*** *Centre for Future Transport and Cities, Coventry University, Coventry, UK (e-mail: ad5271@coventry.ac.uk)*

Abstract: Unstable position and attitude caused by severe motor faults and varying payloads complicates the design of a reliable flight controller for delivery quadcopters during last-mile deliveries. To ensure safe and efficient operation, a novel, robust and adaptive control strategy is employed using a two-stage recursive least squares based estimation, combined with an auto-tuning mixed-sensitivity H_∞ controller. A numerical simulation demonstrates fast and accurate real-time estimation of actuator loss of effectiveness and changes in the mass and inertia due to payload pickup/drop-off during delivery missions. The combined and uneven effects of multiple faults with initial mass and inertia uncertainties are efficiently compensated by this novel fault tolerant flight control strategy. The controller also improves the rejection of these uncertainties during the transient stage before the convergence of the parameter estimation. The proposed approach maintains the operational reliability of delivery quadcopters under a range of challenging uneven faults and significant mass and inertia uncertainties.

Copyright © 2024 The Authors. This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0/>)

Keywords: quadcopter UAV, adaptive control, mixed-sensitivity H_∞ , faults, payload uncertainty

1. INTRODUCTION

The use of unmanned aerial vehicles (UAVs), colloquially known as drones, for payload deliveries is projected to compound with an annual growth rate of 57.5% from 2021 to 2028 (Wirabudi et al. (2022)). Quadcopters, in particular, are applicable for a wide variety of applications, including payload deliveries due to their high agility, precise hovering and vertical takeoff/landing capabilities (Pounds et al. (2010)). A payload delivery quadcopter must account for the changes in mass and inertia due to payload pickup/drop-off during their missions. These delivery quadcopters are required to navigate in tight spaces and crowded urban environments for last-mile deliveries, where fault tolerance becomes unquestionably important to ensure reliable commercial UAV operations. As such, their adoption requires increasingly robust and adaptive fault-tolerant control (FTC).

An uneven drop in the efficiency of the motors, referred to as partial loss of effectiveness (LoE), is a commonly occurring fault during a UAV mission (Puchalski and Giernacki (2022)), and is due to changes in the physical parameters of the motor. An active FTC approach allows for real-time adaptation and precise fault mitigation. There has been significant research on the topic often leading to adaptive control where observers are utilized. A disturbance observer with backstepping control is used by Dong et al.

(2021) to address biases in motors, where the output is shifted from its nominal value. Similarly, the force caused by the biases due to an electric regulator error on the motor is addressed by Ma et al. (2019) using a non-linear high-gain observer with feedback linearization. Motor faults are addressed by Avram et al. (2017) using non-linear adaptive observers, whereas motor failures are addressed by Chung and Son (2020) using observers together with re-configurable control.

Other well-known observers, such as sliding mode observers, unknown input observers, and reduced order observers, have been used in the literature. Among them, recursive least squares (RLS) can be a practical choice for real-time parameter estimation in a quadcopter because of its ability to achieve a high accuracy in a computationally efficient way (Kim et al. (2014)). Moreover, RLS methods allow adaptation to changes in the system over time and are particularly efficient for detecting and isolating actuator faults in a quadcopter (Park et al. (2020)). In Tutsoy et al. (2023), a batch type least squares projection approach was used to quantify the actuator failures estimated using a reduced order Thau observer. Asadi (2022) and Han et al. (2018) used a parity space approach for fault detection and to trigger RLS based actuator fault estimation, whereas Kuric et al. (2017) used RLS for the direct estimation of actuator faults and then reconfigured the control allocation to compensate for them.

Despite the widely recognised practical strengths of least squares based methods, they have remarkably been rarely considered to simultaneously address faults together with other uncertain parameters in multirotor UAVs. To the best of the authors' knowledge, active FTC using least squares methods has never been considered together with quadcopter payload uncertainties previously in the literature. Moreover, the majority of papers using least squares methods also use conventional proportional-integral-derivative (PID) control architectures, which are not very robust to modelling errors and external perturbation. In fact, PID control does not optimise the rejection of external perturbations. Robust H_∞ controllers optimise the rejection of such uncertainties. This is particularly important during the transient time preceding the convergence of RLS or other parameter estimation algorithms. However, a caveat here is that the range of uncertainties that can be handled by the controller is limited and increasing this range leads to a conservative controller with reduced performance (Monfared et al. (2018)). Consequently, a fixed range for quadcopter mass and inertia uncertainties may not guarantee efficient handling of the payload variations for delivery missions.

This paper presents a novel, robust and adaptive control strategy, which uses an auto-tuning mixed-sensitivity H_∞ controller with a two-stage RLS (TSRLS) based estimation to account for faults and payload uncertainties in real-time. The main contributions of this paper are as follows.

- (1) It extends the work in Kuric et al. (2017) to estimate actuator LoE together with mass and inertia using a TSRLS. The first-stage estimates the actuator LoE in all motors and the second-stage estimates the mass and moment of inertia matrix using the non-linear quadcopter dynamics. In doing so, it can account for motor faults together with payload uncertainties (and potentially variations) simultaneously.
- (2) It develops an adaptive control approach based on an auto-tuning mixed-sensitivity H_∞ controller, which takes all estimated parameters from the TSRLS to adapt the controller parameters in real-time. Thus, the controller is robust and adaptive, while avoiding the enforcement of unnecessarily high (conservative) robustness settings.

The remainder of the paper is organized as follows. Section 2 describes the mathematical model of the quadcopter, and Section 3 details the new robust and adaptive control strategy. Section 4 summarizes the performance analysis within numerical simulations before we finish in Section 5 with conclusions and recommendations for future work.

2. MODELLING

The quadcopter is assumed to be a rigid body, which means there is no motion with respect to the body frame. It is symmetrical with an X-configured physical layout as in Fig. 1. Rotors are mounted at the end of each arm, which allows the quadcopter to move and rotate in 3D space.

2.1 Coordinate System

The inertial and body frames of the coordinate system are depicted in Fig. 1. The position vector of the quadcopter

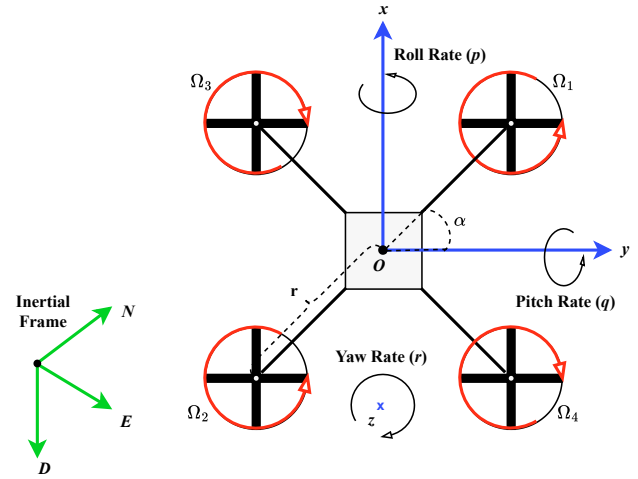


Fig. 1. Quadcopter reference frames and positive rotations

in the inertial frame is denoted as $\xi = [N \ E \ D]^T$ with components along the North, East and Down axes respectively, and its attitude is described by the Euler angles as $\eta = [\phi \ \theta \ \psi]^T$ with components representing the roll, pitch and yaw angles respectively. The equations of motion that describe the changes in position and attitude of the quadcopter are with respect to the inertial frame. However, all forces and moments are expressed within the body frame since the sensors and propellers are attached to it. Euler angles based on the $z-y-x$ rotation sequence are used to describe the rotation matrix, R_B^I and the transfer matrix, T_B^I below, to transform from the body frame to the inertial frame, as explained by Osmić et al. (2016).

$$R_B^I = \begin{bmatrix} c_\theta c_\psi & s_\phi s_\theta c_\psi - c_\phi s_\psi & c_\phi s_\theta c_\psi + s_\phi s_\psi \\ c_\theta s_\psi & s_\phi s_\theta s_\psi + c_\phi c_\psi & c_\phi s_\theta s_\psi - s_\phi c_\psi \\ -s_\theta & s_\phi c_\theta & c_\phi c_\theta \end{bmatrix}$$

$$T_B^I = \begin{bmatrix} 1 & s_\phi t_\theta & c_\phi t_\theta \\ 0 & c_\phi & -s_\phi \\ 0 & s_\phi/c_\theta & c_\phi/c_\theta \end{bmatrix}$$

Here $s_* = \sin(*)$, $c_* = \cos(*)$ and $t_* = \tan(*)$. It is also worth noting that $R_I^B = (R_B^I)^{-1} = (R_B^I)^T$ and can be used to transform from the inertial frame to the body frame. The translational velocities of the quadcopter in the body frame are represented by the vector, $V = [u \ v \ w]^T$ with components along the x , y and z directions respectively, whereas the roll, pitch and yaw rates are represented by the vector, $\omega = [p \ q \ r]^T$ with components about the x , y and z axes respectively. These translational and angular velocities are obtained in the inertial frame using the kinematic equations, $\dot{\xi} = R_B^I V$ and $\dot{\eta} = T_B^I \omega$ respectively.

2.2 Translational and Rotational Dynamics

The dynamics of the quadcopter were derived using the Newton-Euler formalism and the derivations were mainly following the work in Kun and Hwang (2016). The equation relating the forces acting on the quadcopter in the inertial frame is given by

$$mG + R_B^I T = m\ddot{\xi}, \quad (1)$$

where mG and $R_B^I T$ denote the weight and propeller thrust respectively. The quadcopter's mass is denoted by

m , and $\ddot{\xi}$ is its inertial acceleration. The weight acts in the positive D direction. Therefore, mG is expressed as $[0 \ 0 \ mg]^T$, where $g = 9.81$. The thrust vector, T acts perpendicular to the quadcopter in the negative z direction. It is converted to the inertial frame in order to include it in the equations of motion. In other words, propeller thrust is expressed as

$$R_B^I T = - \begin{bmatrix} (c_\phi s_\theta c_\psi + s_\phi s_\psi) T_z \\ (c_\phi s_\theta s_\psi - s_\phi c_\psi) T_z \\ (c_\phi c_\theta) T_z \end{bmatrix}.$$

On the other hand, the equation relating the moments acting on the quadcopter is given in the body frame as

$$\tau = I\dot{\omega} + \omega \times (I\omega), \quad (2)$$

where τ is the propeller torque vector and I is the moment of inertia matrix of the quadcopter. The propeller torque vector acts in the body frame about the centre of gravity (CoG) of the quadcopter and is denoted as $\tau = [\tau_\phi \ \tau_\theta \ \tau_\psi]^T$, where τ_ϕ , τ_θ and τ_ψ represent the rolling, pitching and yawing moments respectively, and are defined positive for clockwise rotations about the x , y and z axes respectively. The gyroscopic moments due to the residual angular velocity of the propellers are not represented here, but in practice they are compensated by the flight controller (Kuric et al. (2017)). The moment of inertia matrix is represented as a diagonal matrix, $I = \text{diag}[I_{xx} \ I_{yy} \ I_{zz}]$, where the diagonal terms denote the inertias about the roll, pitch, and yaw axes respectively.

After rewriting (1) and (2) in component form, the final equations of the translational and rotational dynamics are expressed as follows.

$$\ddot{N} = \frac{1}{m} (-T_z(c_\phi s_\theta c_\psi + s_\phi s_\psi)) \quad (3)$$

$$\ddot{E} = \frac{1}{m} (-T_z(c_\phi s_\theta s_\psi - s_\phi c_\psi)) \quad (4)$$

$$\ddot{D} = g + \frac{1}{m} (-T_z(c_\phi c_\theta)) \quad (5)$$

$$\dot{p} = \frac{1}{I_{xx}} (\tau_\phi - I_{zz}qr + I_{yy}qr) \quad (6)$$

$$\dot{q} = \frac{1}{I_{yy}} (\tau_\theta - I_{xx}pr + I_{zz}pr) \quad (7)$$

$$\dot{r} = \frac{1}{I_{zz}} (\tau_\psi - I_{yy}pq + I_{xx}pq) \quad (8)$$

2.3 Control Inputs, Rotor Dynamics and Motor Mixing

The quadcopter is controlled by adjusting the control inputs, which is done by altering the angular velocity of each rotor through a motor mixing algorithm. Each rotor generates a thrust force, F_i ($i = 1, \dots, 4$), which acts perpendicular to the rotor in the negative z direction, and a torque, τ_i ($i = 1, \dots, 4$) about the CoG of the vehicle.

$$\begin{bmatrix} T_z \\ \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{bmatrix} = \begin{bmatrix} F_1 + F_2 + F_3 + F_4 \\ -F_1 r \cos \alpha + F_2 r \sin \alpha + F_3 r \cos \alpha - F_4 r \sin \alpha \\ F_1 r \sin \alpha - F_2 r \cos \alpha + F_3 r \sin \alpha - F_4 r \cos \alpha \\ -\tau_1 - \tau_2 + \tau_3 + \tau_4 \end{bmatrix}$$

Since the quadcopter is symmetrical and square in shape, the angle between each arm and the y axis, i.e. α as indicated in Fig. 1, is equal to 45° . The rotor dynamics relate the thrust and torques of each rotor to its angular

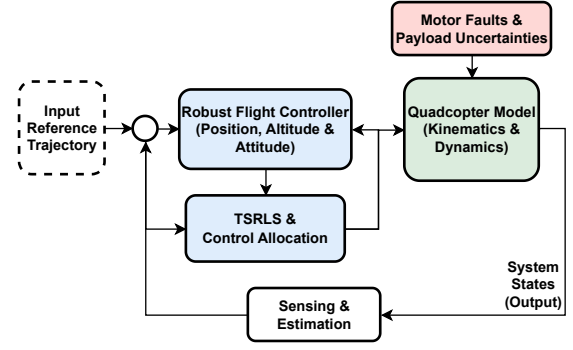


Fig. 2. Adaptive flight control architecture for quadcopter

velocity as $F_i = k_t \Omega_i^2$ and $\tau_i = k_b \Omega_i^2$ respectively, where k_t is the thrust constant and k_b is the drag factor on the propeller. Thus, the actuation model can be expressed using the control inputs and rotor dynamics as follows.

$$\begin{bmatrix} T_z \\ \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{bmatrix} = \begin{bmatrix} k_t & k_t & k_t & k_t \\ -k_t r \sqrt{2}/2 & k_t r \sqrt{2}/2 & k_t r \sqrt{2}/2 & -k_t r \sqrt{2}/2 \\ k_t r \sqrt{2}/2 & -k_t r \sqrt{2}/2 & k_t r \sqrt{2}/2 & -k_t r \sqrt{2}/2 \\ -k_b & -k_b & k_b & k_b \end{bmatrix} \begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{bmatrix}$$

3. ROBUST ADAPTIVE FLIGHT CONTROL

The flight control architecture consists of a cascaded loop, where the altitude and attitude control is performed in an inner-loop and the outer-loop is used for position control. The inner-loop usually works at a higher frequency and generates the thrust and torques required to stabilize the quadcopter. The outer-loop tracks the longitudinal and lateral positions and generates the reference roll and pitch angles. It works at a lower frequency and its performance increases as the inner-loop becomes more robust (Nguyen and Hong (2019)). As such, the inner-loop is the main focus of the contribution, where we implement our novel control strategy, while conventional PID control is used for the outer-loop, which is less associated with actuator LoE.

3.1 Two-Stage RLS

The flight control architecture for a quadcopter is illustrated in Fig. 2. A TSRLS is used to adapt the fault tolerant flight controller to mass and inertia uncertainties. The first-stage estimates the actuator LoE and the second-stage estimates the mass and moment of inertia matrix. These are estimated using the filtered data that comes from an extended Kalman filter (EKF) at each time step by minimizing the sum of the squared errors between the measured and estimated output of the quadcopter as in the weighted least squares cost function:

$$J_i = \sum_{t=1}^n \lambda_i^{n-t} \left(Y_i(t) - X_i(t) \Theta_i(t-1) \right)^2, \quad i = 1, 2.$$

Here i is the stage index and n , λ_i , Y_i , X_i and Θ_i respectively denote the number of samples, forgetting factor, output observation vector, information matrix and the parameters to be estimated. For ease of notation, we dropped the stage index i (e.g., replaced Θ_i by Θ) as both the stages are solved using the same RLS algorithm.

The RLS methodology defined by Isermann and Münchhoff (2011) for least squares parameter estimation was ex-

Table 1. Model parameters

Parameter	Value
radius (r)	0.3 m
mass	1.5* kg
$[I_{xx}, I_{yy}, I_{zz}]$	$[0.02^*, 0.02^*, 0.04^*]$ kgm^2
k_t, k_b	0.0025, 0.00005
* default values without a payload	

Moreover, the mixed-sensitivity H_∞ controller is synthesised using real-time updates of the estimated mass, $\hat{m}$ and estimated inertia, $\hat{I}$ to re-evaluate the controller parameters. Thus, the controller avoids enforcing unnecessarily high (conservative) robustness settings, while maintaining robustness and adaptability to uncertainties.

4. SIMULATION RESULTS

The model parameters used for the simulation and their respective values are given in Table 1. These were approximated with reference to the geometry of the PX4 Vision Autonomy Development Kit (PX4 (2024)), which will be used for experimental validations in future work. The drone delivery scenario involves takeoff with unknown initial payload mass and inertia, and flying a square trajectory to avoid buildings before landing at a different location. Takeoff and landing were not perfectly vertical to provide sufficient information for TSRLS, which is not unusual in a practical setting. The reference values required for the flight controller, i.e. the reference altitude and position coordinates, were generated using a motion planner. The two main aspects of the analysis to validate the adaptive flight controller were as follows.

- (1) Tracking performance of the mixed-sensitivity H_∞ controller with and without TSRLS based estimation.
- (2) Performance comparison between the auto-tuned mixed-sensitivity H_∞ and PID controllers for the inner-loop. Quadcopter inner-loops are indeed typically PID controlled.

The performance enhancement using TSRLS is illustrated in Fig. 4. The H_∞ controller was first tested without the TSRLS based estimation. The altitude, position and attitude stabilization were satisfactory without motor faults or payload uncertainty. A minor motor fault was then injected, where Motor 1 underwent a 1% drop in efficiency. The tracking performance was affected as there was a significant offset in the position. A severe motor fault condition was subsequently considered, where all 4 motors experienced an uneven drop in efficiency with actuator LoE vector, $\Theta_1 = [0.9 \ 0.65 \ 0.5 \ 0.8]^T$. The controller without TSRLS failed to stabilize the quadcopter, which would lead the quadcopter to crash. This test proved the unsuitability of the controller under severe motor faults and the situation worsened in the presence of payload uncertainty. Thus, FTC and adaptability becomes crucial, which was achieved using the TSRLS based estimation.

Next, the controller was tested with the TSRLS based estimation. The forgetting factors of TSRLS were 0.9 and 0.5 for the first and second stages respectively. The same trajectory and the most severe motor faults scenario used previously were considered. Significant payload uncertainty was considered, which makes stabilization even more challenging. The values for mass and inertia are ini-

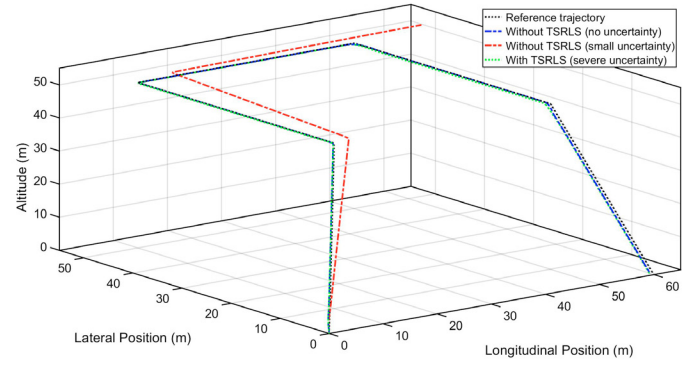


Fig. 4. Tracking performance of mixed-sensitivity H_∞ controller with and without TSRLS based estimation

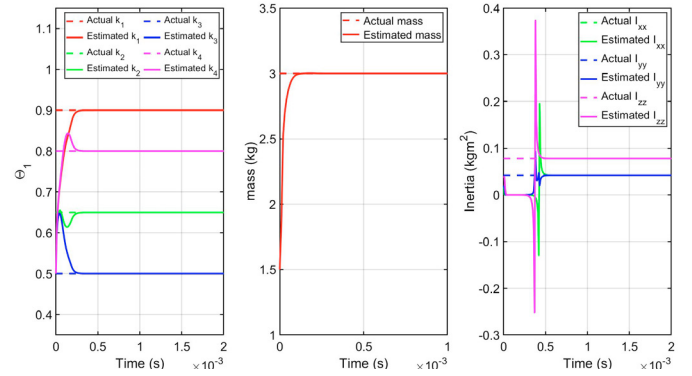


Fig. 5. Actual and estimated values of actuator LoE, mass and inertia for scenario in Fig. 4 – with TSRLS

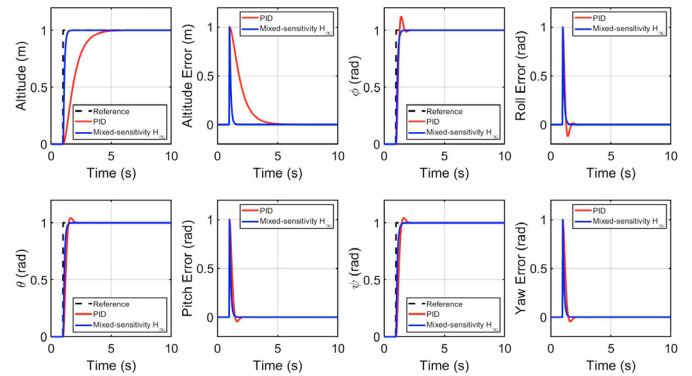


Fig. 6. Performance of the mixed-sensitivity H_∞ and PID controllers

tially unknown in the delivery scenario and were initialised with the default mass and inertia of the quadcopter, i.e. without a payload. The results showed good trajectory tracking performance and the actuator LoE, mass and inertia were accurately estimated in real-time as shown in Fig. 5. This demonstrates that with the proposed TSRLS based estimation, the controller is robust under severe motor faults and significant payload uncertainty where it previously failed without TSRLS based estimation.

A performance comparison between mixed-sensitivity H_∞ and a conventional PID controller architecture was conducted to demonstrate the benefit of the robust controller. Fig. 6 shows the trajectory tracking performance of both controllers. The altitude and attitude stabilization were

satisfactory and were similar to the quadcopter control results presented by Osmić et al. (2016). However, the response time is much better for mixed-sensitivity H_∞ than the PID controller architecture. This is particularly important because of the ability of robust control to rapidly reject destabilising disturbance effects compared to the PID controller architecture, especially before the convergence of TSRLS. Moreover, the PID architecture fails with high initial payload uncertainty as it is tuned for the default mass and inertia, and the online tuning of PID gains for quadcopters is complex as the PID architecture typically employs 6 individual PID controllers, each one coupled.

5. CONCLUSION

A new robust adaptive quadcopter control architecture was proposed to address complex stability issues under multiple uneven motor faults and significant mass and inertia uncertainties. The approach employs an auto-tuning mixed-sensitivity H_∞ controller alongside a TSRLS to accurately estimate motor faults, mass and inertia. These estimates inform an adaptive mechanism that fine-tunes individual rotor speeds and re-evaluates the controller parameters, thereby ensuring optimal controller performance while avoiding unnecessarily high (conservative) robustness settings. The simulation results demonstrate accurate position and attitude control under severe motor faults and significant payload uncertainty, which enhances operational safety for a drone based payload delivery mission.

Future work will include enhancing controller resilience to additional factors, including wind disturbance, and the practical validation of the proposed solutions.

REFERENCES

- Asadi, D. (2022). Model-based fault detection and identification of a quadrotor with rotor fault. *International Journal of Aeronautical and Space Sciences*, 23(5), 916–928. doi:10.1007/s42405-022-00494-z.
- Avram, R.C., Zhang, X., and Muse, J. (2017). Quadrotor actuator fault diagnosis and accommodation using nonlinear adaptive estimators. *IEEE Transactions on Control Systems Technology*, 25(6), 2219–2226. doi:10.1109/TCST.2016.2640941.
- Başak, H. and Prempain, E. (2014). Low order H_∞ controllers for a quadrotor UAV. In *2014 UKACC International Conference on Control*, 127–132. doi:10.1109/CONTROL.2014.6915127.
- Chung, W. and Son, H. (2020). Fault-tolerant control of multirotor UAVs by control variable elimination. *IEEE/ASME Transactions on Mechatronics*, 25(5), 2513–2522. doi:10.1109/TMECH.2020.2982436.
- Dong, G., Cao, L., Yao, D., Li, H., and Lu, R. (2021). Adaptive attitude control for multi-MUAV systems with output dead-zone and actuator fault. *IEEE/CAA Journal of Automatica Sinica*, 8(9), 1567–1575. doi:10.1109/JAS.2020.1003605.
- Han, W., Wang, Z., and Shen, Y. (2018). Fault estimation for a quadrotor UAV by integrating the parity space approach with RLS. *Proceedings of the IMechE - Part G*, 232(4), 783–796. doi:10.1177/0954410017691794.
- Isermann, R. and Münchhof, M. (2011). Identification of dynamic systems. Chapter 9. Springer, Berlin.
- Kim, T., Wang, Y., Sahinoglu, Z., Wada, T., Hara, S., and Qiao, W. (2014). Fast UD factorization-based RLS online parameter identification for model-based condition monitoring of lithium-ion batteries. In *2014 American Control Conference*, 4410–4415. doi:10.1109/ACC.2014.6859108.
- Kun, D.W. and Hwang, I. (2016). Linear matrix inequality-based nonlinear adaptive robust control of quadrotor. *Journal of Guidance, Control, and Dynamics*, 39(5), 996–1008. doi:10.2514/1.G001439.
- Kuric, M., Lacevic, B., Osmic, N., and Tahirovic, A. (2017). RLS-based fault-tolerant tracking control of multirotor aerial vehicles. In *2017 IEEE International Conference on Advanced Intelligent Mechatronics*, 1148–1153. doi:10.1109/AIM.2017.8014173.
- Ma, H.J., Liu, Y., Li, T., and Yang, G.H. (2019). Nonlinear high-gain observer-based diagnosis and compensation for actuator and sensor faults in a quadrotor unmanned aerial vehicle. *IEEE Transactions on Industrial Informatics*, 15(1), 550–562. doi:10.1109/TII.2018.2865522.
- Monfared, S.B., Kalhor, A., and Atashgah, M.A. (2018). Robust adaptive H_∞ control for attitude control of a quadrotor. In *6th RSI International Conference on Robotics and Mechatronics*, 560–565. doi:10.1109/ICRoM.2018.8657497.
- Nguyen, N.P. and Hong, S.K. (2019). Fault-tolerant control of quadcopter UAVs using robust adaptive sliding mode approach. *Energies*, 12(1). doi:10.3390/en12010095.
- Osmić, N., Kurić, M., and Petrović, I. (2016). Detailed octorotor modeling and PD control. In *2016 IEEE International Conference on Systems, Man, and Cybernetics*, 002182–002189. doi:10.1109/SMC.2016.7844562.
- Park, J.H., Jun, C.Y., Jeong, J.Y., and Chang, D.E. (2020). Real-time quadrotor actuator fault detection and isolation using multivariate statistical analysis techniques with sensor measurements. In *2020 20th International Conference on Control, Automation and Systems*, 33–37. doi:10.23919/ICCAS50221.2020.9268391.
- Pounds, P., Mahony, R., and Corke, P. (2010). Modelling and control of a large quadrotor robot. *Control Engineering Practice - Special Issue on Aerial Robotics*, 18(7), 691–699. doi:10.1016/j.conengprac.2010.02.008.
- Puchalski, R. and Giernacki, W. (2022). UAV fault detection methods, state-of-the-art. *Drones*, 6(11). doi:10.3390/drones6110330.
- PX4 (2024). PX4 vision autonomy development kit. URL https://docs.px4.io/main/en/complete_vehicles_mc/px4_vision_kit.
- Skogestad, S. and Postlethwaite, I. (2005). *Multivariable Feedback Control: Analysis and Design*. John Wiley & Sons, Inc., Hoboken, NJ, USA.
- Tutsoy, O., Asadi, D., Ahmadi, K., and Nabavi-Chasmi, S.Y. (2023). Robust reduced order thau observer with the adaptive fault estimator for the unmanned air vehicles. *IEEE Transactions on Vehicular Technology*, 72(2), 1601–1610. doi:10.1109/TVT.2022.3214479.
- Wirabudi, A.A., Hafiza, L., and Fachrurrozi, N.R. (2022). Design autonomous drone control for delivery package using prim algorithm and waypoint method. In *13th International Conference on Information and Communication Technology Convergence*, 1183–1188. doi:10.1109/ICTC55196.2022.9952874.